Module 1:
Snell's Laws

Course 1 of *Optical Engineering*: First Order Optical System Design

with **Dr. Robert R. McLeod and Dr. Amy C. Sullivan**

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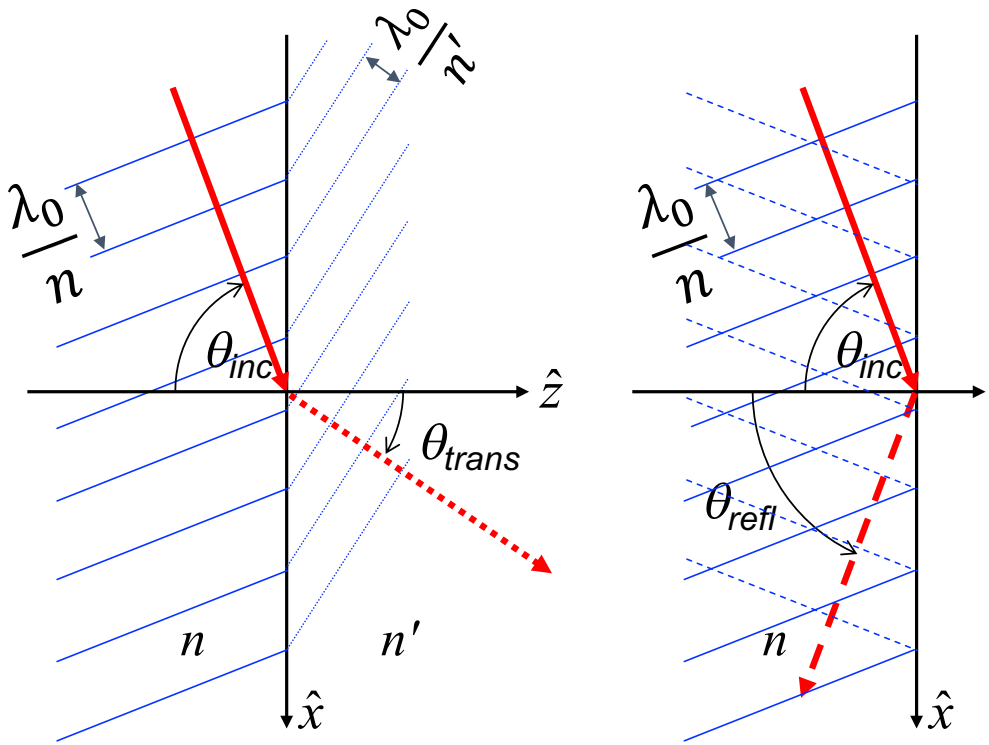
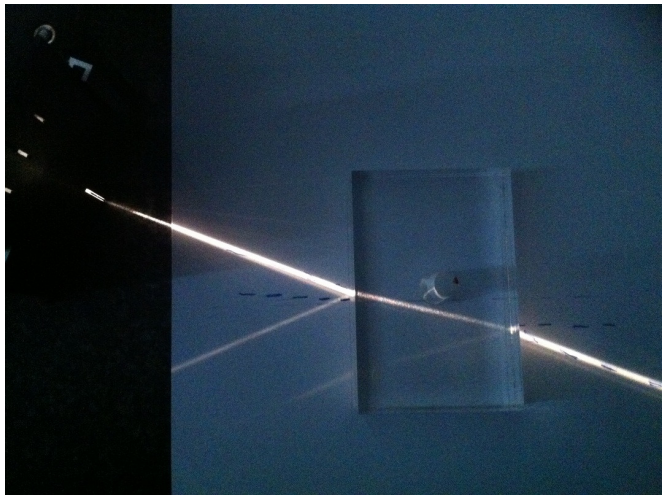
Snell's laws: How rays change at surfaces

$$n \sin\theta_{inc} = n' \sin\theta_{trans}$$

$$\theta_{inc} = \theta_{refl}$$



- Transmission depends on n, n'
- Reflection does not
- Both can be derived from continuity of E



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Module 1:
Fermat's Principle

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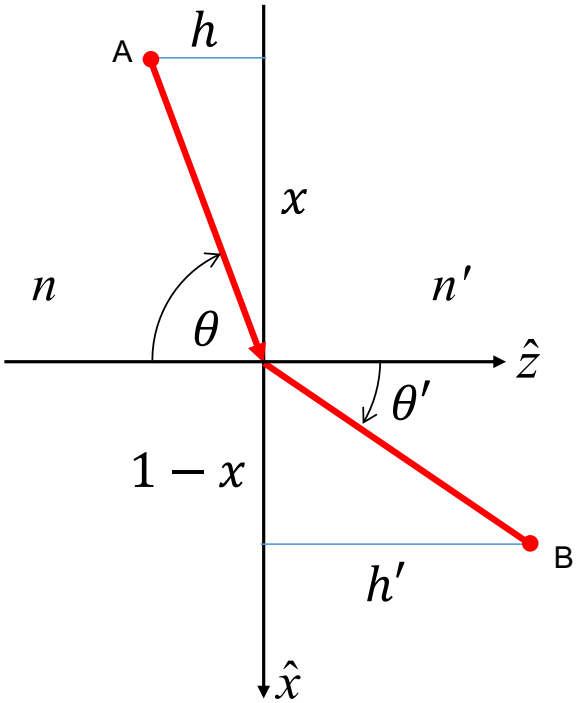
Fermat's principle: or the principle of least time

Consider a ray travelling from A to B



Fermat's principle states that rays traverse the path of stationary optical length with respect to variations of the path. Typically this means that rays take the path that requires the least travel time. The OPL from A to B is

$$S_{AB} = n\sqrt{x^2 + h^2} + n'\sqrt{(1-x)^2 + h'^2}$$

Let S be stationary with respect to x

$$0 = \frac{d}{dx} S_{AB} = n\frac{x}{\sqrt{x^2 + h^2}} + n'\frac{-(1-x)}{\sqrt{x^2 + h'^2}}$$

$$\boxed{= n\sin\theta - n'\sin\theta'}$$

Thus Fermat's principle is equivalent to Snell's Law.

This can be cast as the "lifeguard problem" in which a lifeguard at A wishes to most quickly reach a swimmer at B.

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